not reveal how these values are dispersed (spread or scattered) on each side of the central value. Therefore while describing data set it is equally important to know how for the item in the data are close around or scattered away from the measures of central tendency.

Example 6.1

Look at the runs scored by the two cricket players in a test match:

Players	I Innings	II Innings	Mean
Player 1	0	100	50
Player 2	40	60	50

Comparing the averages of the two players we may come to the conclusion that they were playing alike. But player 1 scored 0 runs in I innings and 100 in II innings. Player 2 scored nearly equal runs in both the innings. Therefore it is necessary for us to understand data by measuring dispersion.

6.1 Characteristics of a good Measure of Dispersion

An ideal measure of dispersion is to satisfy the following characteristics.

(i) It should be well defined without any ambiguity.

(ii) It should be based on all observations in the data set..

(iii) It should be easy to understand and compute.

(iv) It should be capable of further mathematical treatment.

(v) It should not be affected by fluctuations of sampling.

(vi) It should not be affected by extreme observations.

6.2 Types of measures of dispersion

Range, Quartile deviation, Mean deviations, Standard deviation and their Relative measures

The measures of dispersion are classified in two categories, namely

(i) Absolute measures

(ii) Relative measures.

6.3 Absolute Measures

It involves the units of measurements of the observations. For example, (i) the dispersion of salary of employees is expressed in rupees, and (ii) the variation of time

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